

Norton's Theorem

Actual Statement of Theorem

Norton's Theorem states that any complex, linear, two-terminal circuit can be replaced by an equivalent circuit consisting of a single current source (I_N) in parallel with a single resistance (R_N) connected to a load.

This equivalent circuit produces the same current and voltage at the load terminals as the original circuit.

$$I = (I_N \times R_N) / (R_N + R_L)$$

This formula gives the current flowing through the load resistor. It shows that the load current depends on the Norton current and the total resistance in the circuit.

Relation to Thevenin's Theorem

Norton's theorem is closely related to Thevenin's theorem. Both represent the same circuit in different forms:

- Thevenin: Voltage source in series with resistance
- Norton: Current source in parallel with resistance

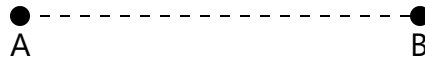
$$I_N = V_{th} / R_{th} \quad R_N = R_{th}$$

This means any Thevenin equivalent can be directly converted into a Norton equivalent.

Step-by-Step Procedure

Step 1: Disconnect the Load

Remove the load resistor and identify the two output terminals. These are the points across which the Norton equivalent will be found.



Step 2: Find Norton Resistance (R_N)

Deactivate all independent sources:

- Voltage sources → Short circuit
- Current sources → Open circuit

Then calculate the equivalent resistance seen from the terminals.

$$R_N = R_{eq}$$

Step 3: Find Norton Current (I_N)

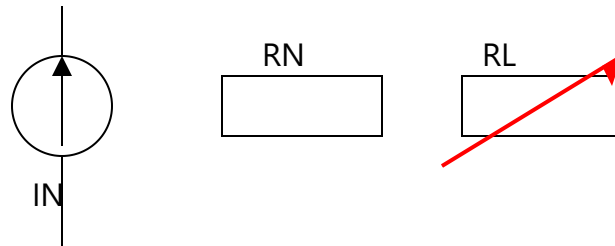
The Norton current is the short-circuit current between the two terminals.

$$I_N = I_{sc}$$

This can be calculated using nodal analysis, mesh analysis, or Ohm's Law depending on the circuit.

Step 4: Construct Norton Equivalent Circuit

The final circuit consists of a current source in parallel with R_N , connected to the load resistor.



The arrow across the load resistor represents a **variable resistor**, indicating that its value can change.

Methods to Solve the Circuit

1. Ohm's Law

Used when the circuit is simple.

$$I = V / R$$

2. Nodal Analysis

Apply Kirchhoff's Current Law at nodes. **Use when:** Many parallel branches.

3. Mesh Analysis

Apply Kirchhoff's Voltage Law in loops. **Use when:** Few loops.

4. Source Transformation

Convert between Thevenin and Norton forms. **Use when:** Simplifying circuits.

5. Superposition

Analyze each source separately. **Use when:** Multiple independent sources exist.

Practical Applications

Norton's theorem is widely used in electrical engineering to simplify circuits, especially those involving current sources.

It is particularly useful in analyzing parallel networks, current division problems, and power systems.

Engineers also use Norton equivalents when designing circuits that must deliver specific current levels to loads.